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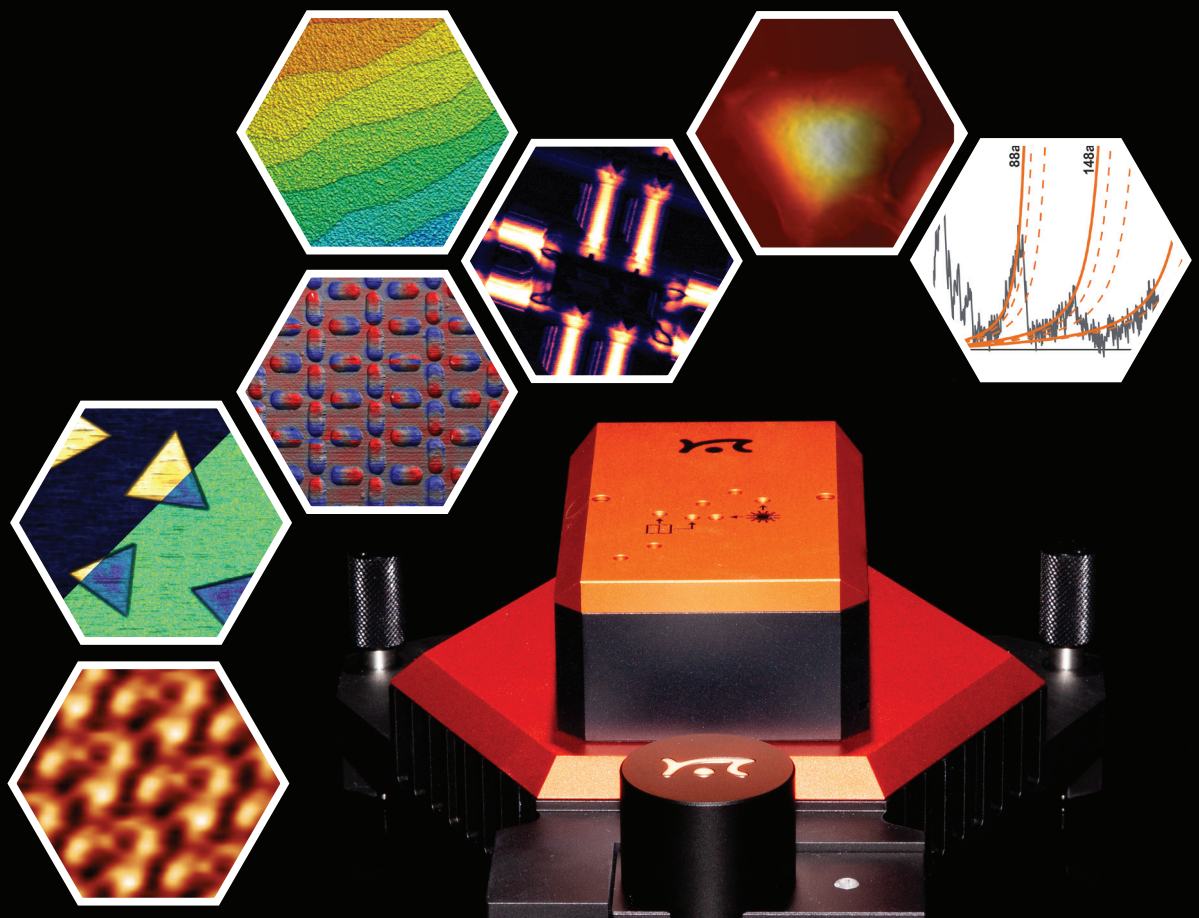
FlexAFM

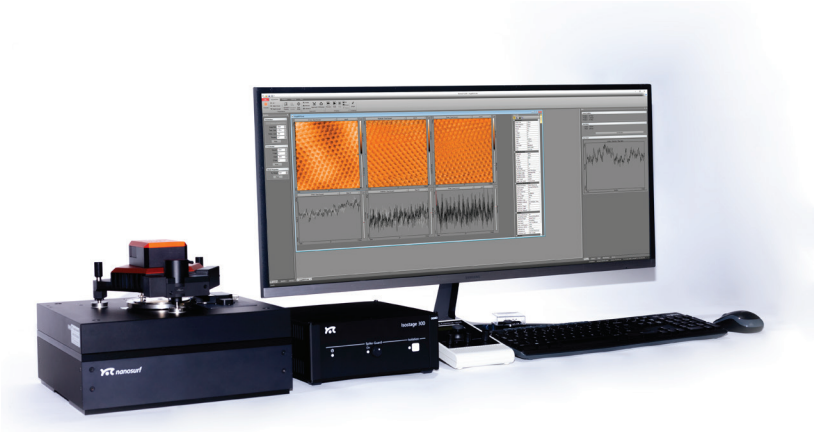
The Flexible Research AFM



The Flexible Research AFM

The FlexAFM has been the instrument of choice for hundreds of researchers in need of a reliable and versatile atomic force microscope system. Built with ease of use and configurability in mind, the capabilities of the FlexAFM cover most research needs: Configure your FlexAFM for 2D materials research in a glovebox, or integrate it with your inverted optical microscope for biological applications - no matter what your research goals are, the Nanosurf FlexAFM provides you with an affordable mid-range instrument that you can expand and upgrade as your requirements evolve.

Why FlexAFM?



- Reliable and robust: flexure-based scanner provides flat and linear scanning
- Versatile: flexible sample size and weight thanks to tip-scanning technology
- Configurable to your needs: from basic research to advanced experiments
  - Electrical modes
  - Spectroscopy
  - Motorization and automation
  - Inverted optical microscope integration for bio-research setup and FluidFM
  - Turnkey Glovebox solution available
- Choice of C3000i or CX Controller for additional user input/output channels
- Future-proof: shared accessories universe with DriveAFM for easy upgrade to high-end AFM
- User-friendly handling, light and robust design

Individual Configuration

A large selection of accessories and functional add-on options are available to expand your FlexAFM's capabilities. Choose from a variety of sample holders and stages to tailor suit your needs: whether you are performing biological research on an inverted optical microscope setup or engaged in materials science with the need for the stringent environmental control of a glovebox, the FlexAFM covers all bases.

From Electrical Modes to Live Cell Research

Functional sample holders and add-ons pave the road for experiments using electrical modes, including scanning microwave microscopy, sample heating or cooling, electrochemistry, even a live cell incubator. You have all the application flexibility you need.

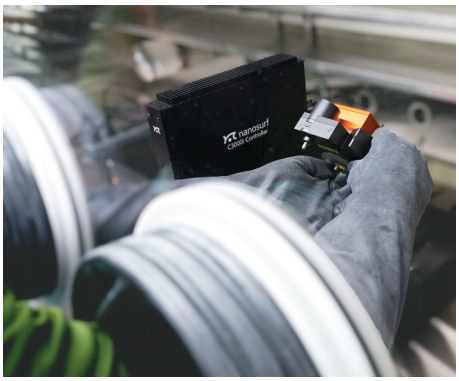
Budget Effectiveness

Thanks to the highly flexible design, you will get the most out of your available budget: our technical sales team has helped hundreds of customers configure their FlexAFM to fulfil all their needs despite limited budgets.

Seamless Upgrade Path

As your research evolves, you have the option for a phased upgrade from FlexAFM to DriveAFM. Both systems share a common footprint, ensuring that your existing accessories, stages, and sample holders remain fully functional. Start by upgrading to the CX controller to give your FlexAFM a performance boost. When you're ready for even more advanced capabilities, proceed to upgrade the scan head to DriveAFM. This modular approach not only enhances performance but also offers significant cost savings.

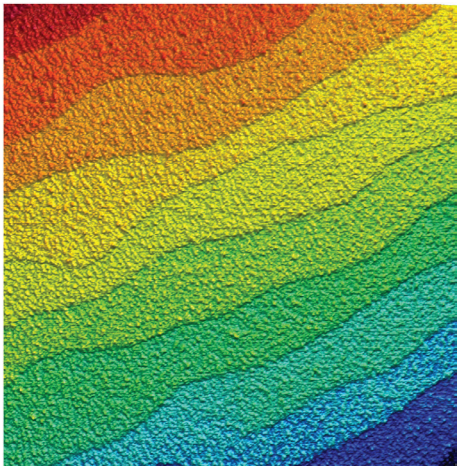
Stand-alone Setup



Glovebox-ready

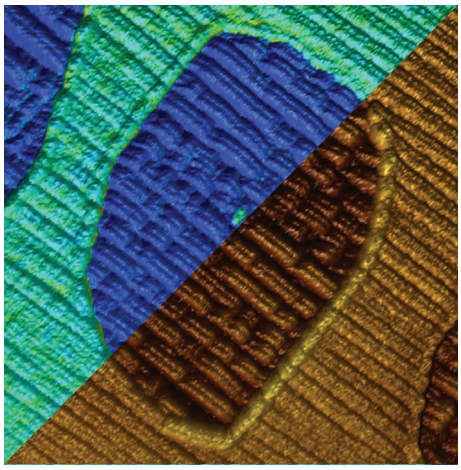
Optical Microscope Setup

Topography in Air



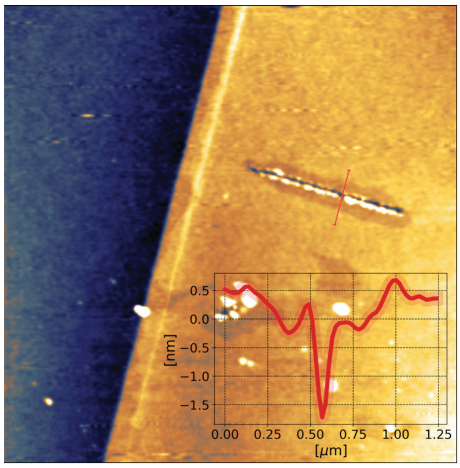
Topography of $SrTiO_3$ in dynamic mode. Scan size: 1 x 1 µm².

Graphene



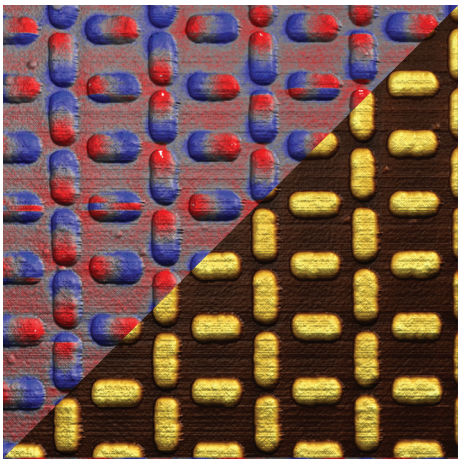
Quality control of CVD-grown graphene flakes on post-oxidized copper measured by friction (data courtesy: Newtec engineering A/S). Scan size: 10 x 10 µm²

Graphene Cutting



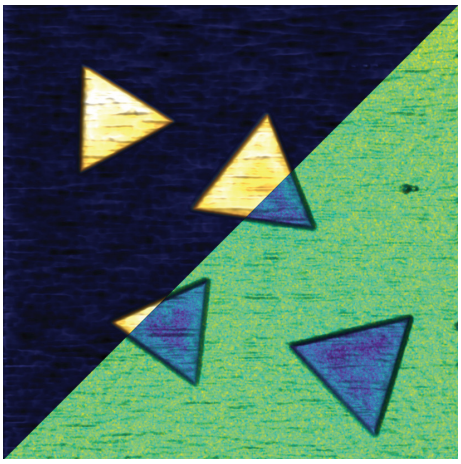
Cutting graphene by AFM lithography. Sample courtesy: Kim group, Harvard University, USA.

MFM



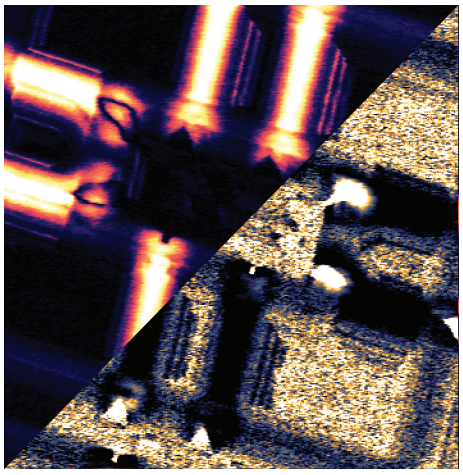
MFM and topography on artificial spin ice (data courtesy: Prof. S. Ferreira). Scan size: 4 x 4 µm².

KPFM



Topography (left) and KPFM signal (right) of CVD grown MoS2 crystals. Scan size: 42 x 42 µm²; Z-range: 9 nm; KPFM range: 1.45 V.

SMM



Amplitude (left) and phase (right) of dS/dV (dC/dV) measurement on SRAM sample. Scan size: 11 x 11 µm².

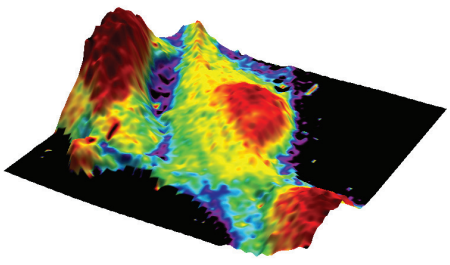
Spectroscopy and Force Mapping

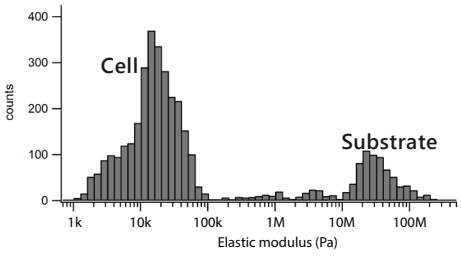
Designed to meet the demanding requirements of scientific research, the FlexAFM excels in force spectroscopy applications.

It is a powerful tool to record force curves, either with the internal z-piezo or with an external piezo that increases the z-range up to 150 µm.

The FlexAFM also includes voltage spectroscopy functionality, adding another layer of analytical power to its feature set.

To facilitate comprehensive force curve analysis, we offer the ANA offline software, complete with histogram creation capabilities, at no additional cost.



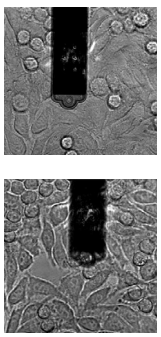


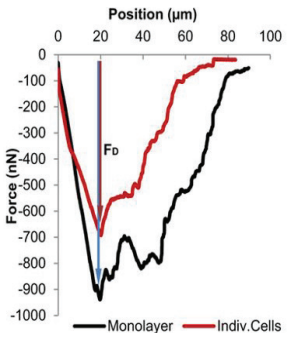
Left: Elastic modulus mapped onto 3D topopgraphy recorded on living breast epithelial cells immersed in cell culture medium.

Right: Elastic modulus distribution.

FluidFM

FluidFM brings microfluidics to the tip of an AFM cantilever. It combines microfluidics with the force sensitivity and positional accuracy of an AFM, paving the path for a range of exciting applications in single-cell biology and nanoscience.





Left: Optical guidance helps to select single cells that are isolated (top) or in a confluent layer (bottom).

Right: Single cell force curves depicting the increase in adhesion force by cell-cell interactions [Scientific Reports 7, 46152].

FlexAFM Technical Specifications

FlexAFM 5 Scan Head Specifications

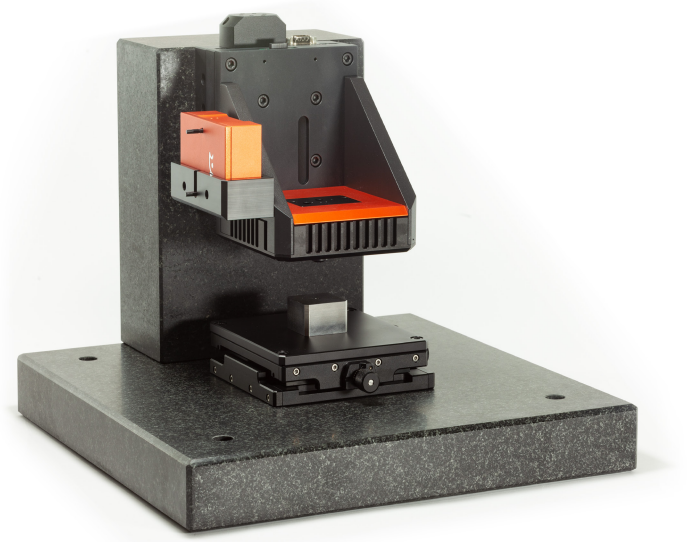
Scan head type	100-µm	10-µm
Sample size	Unlimited w/o sample stage 100 mm on sample stage	
Manual height adjustment range	6 mm	
Motorized approach range (at tip position)	2 mm	
Maximum scan range	100 µm (1)	10 µm (1)
Maximum Z-range	10 µm (2)	3 µm (1)
XY-linearity mean error	< 0.1%	
XY-flatness at maximum scan range	typ. 5 nm	typ. 1 nm
Detector bandwidth	DC – 4 MHz	
Z-sensor noise level (RMS)	typ. 180 pm / max. 200 pm (3)	
Z-measurement noise level (RMS, static mode in air)	typ. 100 pm / max. 200 pm	
Z-measurement noise level (RMS, dynamic mode in air)	typ. 35 pm / max. 50 pm	

(1) Manufacturing tolerances ± 5%

(2) Manufacturing tolerances ± 10%

(3) Measured at 2 kHz

	C3000i Controller	CX Controller
X/Y/Z-axis scan and position controller	3× 24-bit DAC (200 kHz)	3x 28-bit DAC (1 MHz)
X/Y/Z-axis position measurement	1× 24-bit ADC (200 kHz)	3x 20-bit ACD (1 MHz)
Main input signal capturing	2× 16-bit ADC (20 MHz) 2× 24-bit ADC (200 kHz)	2× 16-bit ADC (100 MHz) 8× 20-bit ADC (1 MHz)
User inputs	1× 24-bit ADC (200 kHz)	1× 16-bit ADC (100 MHz) 4× 20-bit ADC (1 MHz)
User outputs	1x 24-bit DAC (200 kHz)	1× 16-bit DAC (100 MHz) 4× 28-bit DAC (1 MHz)
Signal analyzers	2 signal analyzer function blocks that can be configured as dual channel lock-in	2 signal analyzer function blocks that can be configured as dual channel lock-in
Digital synchronization	Sync Out 1/2: 0/5V TTL pulses	2-bit sync out 5 V/TTL galvanically isolated, Spike-Guard input
Scripting	Scripting through COM interface Python library available	Scripting through COM/LUA interface Python library available
Communication	USB 2.0 high-speed to PC	GB Ethernet to PC



Example setup: Flex-Mount on standard 400 mm XYZ-stage

Flex-Mount: FlexAFM precision in a mountable format

The Flex-Mount solution combines the superior resolution and performance of the Nanosurf FlexAFM scan head with the mountability of the Nanosurf NaniteAFM.

The result is an AFM that can easily be attached to one of Nanosurf's large or custom-built automated translation stages, and is thus able to characterize the surface of samples with weights and dimensions virtually no other AFM system can handle.

It is the ultimate quality control instrument to acquire high-resolution information on large and demanding samples.

Scan head specifications

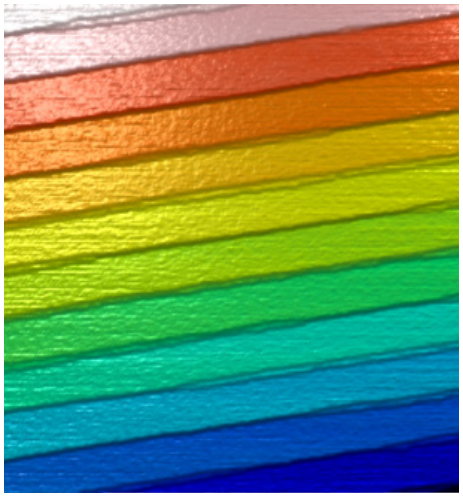
Flex-Mount scan head features

General design	Mountable scan head with tip scanner; Flexure-based electro-magnetically actuated XY-scanner with superb linearity; Piezo-based Z-actuator; Optical Z-position sensor; Closed loop Z-control
Laser / detector	High-speed, low-noise 4-quadrant photodiode detector; near-infrared SLD; laser on/off through software and scan head tilting; optical filters for use with optical microscope phase contrast and fluorescence
Approach	50 mm z-stage with step-by-step approach; software-driven motorized coarse movement and automated final approach
Cantilever holder	Automatic self-alignment for cantilevers with alignment grooves. Manual laser adjustment possible for special cantilevers.
Sample observation	Top and side view in air and liquid; White LEDs (brightness 0–100%); Axial illumination for top view
Operating modes	Static Force, Lateral Force, Dynamic Force, Phase Contrast, MFM, EFM, KPFM, Piezo Force, Force Modulation, Scanning Thermal, Spreading Resistance, Multiple Spectroscopy modes, Lithography and Manipulation modes. Some modes may require additional hardware and/or activating of the respective C3000 controller options.
Security out option (for stage controllers)	Security mechanism to prevent severe scan head crash and to protect the sample. Includes table-top operating unit with emergency kill switch and manual security override for 3rd party stage control units by LSTEP or SmarAct.

Flex-Mount scan head with C3000i controller

Scan head type	100-µm
Sample size	Stage-dependent
Motorized approach range (at tip position with standard z-stage)	50 mm
Maximum scan range	100 µm (1)
Maximum Z-range	10 µm (2)
XY-linearity mean error	< 0.1%
XY-flatness at maximum scan range	typ. 5 nm
Detector bandwith	DC – 4 MHz
Detector noise level	typ. 60 pm / max. 100 pm (3,4)
Z-sensor noise level (RMS)	typ. 180 pm / max. 200 pm (3)
Z-measurement noise level (RMS, static mode in air)	typ. 100 pm / max. 200 pm
Z-measurement noise level (RMS, dynamic mode in air)	typ. 35 pm / max. 50 pm
Scan head dimensions incl. z-stage	181 mm × 130 mm × 171 mm ±25 mm
Scan head weight incl. z-stage	4.7 kg

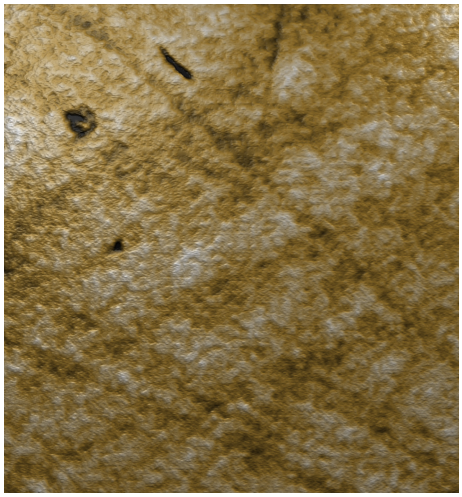
(1) Manufacturing tolerances ± 5%
(2) Manufacturing tolerances ± 10%
(3) Measured at 2 kHz
(4) Measured with XYContr cantilever



SiC steps

Scan size:
1.5 µm x 1.5 µm

The scan shows step heights of 1.5 nm between the large terraces, and 0.75 nm between the large and small terraces' half-steps.



Glass

Scan size:
5 µm x 5 µm

Surface roughness:
0.112 nm RMS
(0.087 nm Ra)

The leading manufacturer of Atomic Force and Scanning tunneling Microscopes for Nano-research

AFM for Materials Science Research

Drive AFM

A fully automated AFM for top notch research with a lot of first time feature such as clean drive, wave mode etc. Compatible with manual and motorized XY stages of different dimensions.

https://www.nanosurf.com/en/products/driveafm

Flex AFM

A flexible research grade AFM with the option of customization as per customer requirement. Compatible with manual and motorized XY stages of different dimensions.

https://www.nanosurf.com/en/products/flexafm

AFM for Biological Science Research

Drive AFM with Inverted Optical Microscope

The fully automated system capable of integration with inverted optical microscopes to perform dual imaging including AFM/ Fluorescence, AFM/Confocal and many more.

https://www.nanosurf.com/en/products/driveafm#life-science

Flex AFM with Inverted Optical Microscope

A versatile and flexible AFM capable of integration with inverted optical microscopes to perform dual imaging including AFM/ Fluorescence, AFM/Confocal and many more.

https://www.nanosurf.com/en/products/flexafm#life-science

Core AFM with Digital Inverted Optical Microscope

Budget friendly AFM with digital inverted microscope (DIMO) option to perform dual imaging without the need for stand alone inverted microscope.

https://www.nanosurf.com/en/products/coreafm-the-essence-of-atomic-force-microscopy#dimo

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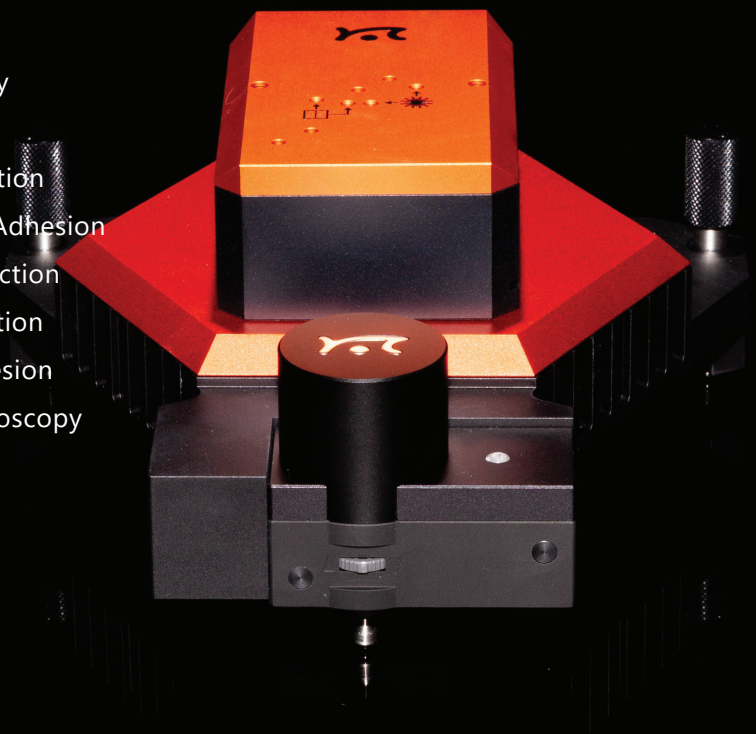
- Electrochemistry
- Signal I/O
- Scanning Thermal Microscopy
- Environmental Control
- Top & Side-View Camera
- Relative Humidity
- Inverted Microscope Stage
- Motorized Inverted Microscope Stage
- Petri Dish
- Petri Dish Heating
- 100 µm Z-Stage
- Digital Inverted Microscope
- Coverslip Holder
- Advanced Optics
- FluidFM®
  - Spotting
  - Nanolithography
  - SICM
  - Single Cell Injection
  - Single Bacteria Adhesion
  - Single Cell Extraction
  - Single Cell Isolation
  - Single Cell Adhesion
  - Colloidal Spectroscopy
- Advanced Spectroscopy
- ANA Add-On
- Automation
- Motorized Translation Stage
- Scripting Interface
- Acoustic Enclosure
- Variable Magnetic Field
- Advanced Lithography
- Contour Following Mode
- Heater/Cooler
- Conductive AFM
- Advanced Conductive AFM
- PFM Mode
- EFM Mode
- KPFM Mode



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